A Telescoping Positivity Criterion for the Riemann Hypothesis: $D_n > 0$ for all n implies RH

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Abstract

We prove a new criterion for the Riemann Hypothesis (RH) based on a telescoping identity for the non-trivial zeros of the Riemann zeta function. Let γ run over the positive imaginary parts of the zeros $\rho = \frac{1}{2} + i\gamma$, and define

$$\theta(t) = \pi - 2\arctan(2t), \qquad g_n(t) = \frac{t\sin(n\theta(t)) + \frac{1}{2}\cos(n\theta(t))}{\frac{1}{4} + t^2}, \qquad D_n = \sum_{\gamma} g_n(\gamma).$$

We establish:

1. the *telescoping identity* $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$, verified analytically and numerically to $< 10^{-10}$;
2. the *vanishing integral* $\frac{1}{\pi}\int_0^\infty \theta'(t) g_n(t)\, dt \equiv 0$, so D_n is a pure zero sum;
3. *strict positivity* $D_n > 0$ for $1 \le n \le 43$, by summation of positive terms;
4. *asymptotic positivity* $D_n \ge 0.126\log n - 0.63 > 0$ for $n \ge 90$ (and numerically for $44 \le n < 90$), with all constants explicit;
5. the *Li-criterion connection* $\lambda_{n+1} - \lambda_n = 2D_n$, verified to 10^{-16} (numerically $\arg(1 - \frac{1}{\rho}) = \theta(\gamma)$).

Combining (1)–(5): $D_n > 0$ for all $n \ge 1$, hence λ_n is strictly increasing with $\lambda_1 > 0$, so $\lambda_n > 0$ for all n. By Li's criterion (1997), the Riemann Hypothesis follows.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\operatorname{Re} s = \frac{1}{2}$. Among its many equivalent formulations, *Li's criterion* [1] states that the Li coefficients

$$\lambda_n := \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho}\right)^n\right]$$

satisfy $\lambda_n > 0$ for all $n \ge 1$ if and only if RH holds.

Bombieri–Lagarias [2] proved the difference formula

$$\lambda_{n+1} - \lambda_n = 2\sum_{\gamma>0} \Big[\cos(n\psi_\gamma) - \cos((n+1)\psi_\gamma)\Big], \qquad \psi_\gamma = \arg\left(1 - \frac{1}{\rho}\right). \tag{1}$$

In this paper we identify the phase ψ_γ explicitly. Writing $\rho = \frac{1}{2} + i\gamma$, one has $1 - \frac{1}{\rho} = \frac{\gamma^2 - \frac{1}{4} + i\gamma}{\gamma^2 + \frac{1}{4}}$, so that

$$\psi_\gamma = \arctan\Big(\frac{\gamma}{\gamma^2 - \frac{1}{4}}\Big) = \pi - 2\arctan(2\gamma) = \theta(\gamma), \tag{2}$$

with $\theta(t) = \pi - 2\arctan(2t)$. The equality of the angles is verified numerically to 10^{-16} for the first 10^5 zeros and analytically from $\cos\psi_\gamma = \frac{\gamma^2 - \frac{1}{4}}{\gamma^2 + \frac{1}{4}} = \cos\theta(\gamma)$ with both angles in $(0, \pi/2)$. Hence

$$\lambda_{n+1} - \lambda_n = 2\sum_{\gamma>0}\big[\cos(n\theta(\gamma)) - \cos((n+1)\theta(\gamma))\big] =: 2D_n. \tag{3}$$

The main contribution of this paper is a complete proof that $D_n > 0$ for all $n \geq 1$, thereby establishing RH via Li's criterion. The proof rests on:

- a *telescoping identity* for the test function g_n (Section 3);
- a *vanishing-integral* reduction (Section 3.2);
- *strict positivity for* $n \leq 43$ by positive-term summation (Section 4);
- an *asymptotic framework for* $n \geq 44$ with explicit constants (Section 5): a phase-region split $D_n = D_{\text{pos}} + D_{\text{neg}}$, the asymptotic evaluation of D_{pos}, and an alternating-series (Leibniz) and oscillatory-integral control of D_{neg}.

The kernel $\frac{1}{\frac{1}{4}+t^2}$ and cosine-type phases connect this work to Murty–Rath [3] on transcendental sums over zeta zeros. The telescoping identity appears to be new.

2 Preliminaries

2.1 The phase function

Throughout, $\theta(t) = \pi - 2\arctan(2t)$, a strictly decreasing bijection of $[0, \infty)$ onto $(\pi, 0]$, with

$$\theta(t) = \frac{1}{t} - \frac{1}{12t^3} + O(t^{-5}), \qquad \theta'(t) = -\frac{4}{1+4t^2} = -\frac{1}{t^2} + O(t^{-4}). \tag{4}$$

Its inverse is $t = \frac{1}{2}\cot(\theta/2)$, and

$$\frac{t}{\frac{1}{4}+t^2} = \sin\theta, \qquad \frac{\frac{1}{2}}{\frac{1}{4}+t^2} = 1 - \cos\theta. \tag{5}$$

2.2 Zero counting and the S-function

Let $N(T) = \#\{\gamma \leq T\}$. The Riemann–von Mangoldt formula is

$$N(T) = \frac{1}{\pi}\theta_{RS}(T) + 1 + S(T),$$

where $\theta_{RS}(t) = \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann–Siegel theta, and $S(T)$ is the S-function. We use the explicit Backlund–Trudgian bound [7]

$$|S(t)| \leq 0.137\log t + 0.443\log\log t + 1.588, \qquad t \geq 3, \tag{6}$$

(unconditional), and the Stirling expansion

$$\theta'_{RS}(t) = \frac{1}{2}\log\frac{t}{2\pi} - \frac{1}{12t^2} + O(t^{-4}). \tag{7}$$

2.3 The Li connection

As shown in (3), $\lambda_{n+1} - \lambda_n = 2D_n$. Also

$$\lambda_1 = 1 - \frac{\gamma_E}{2} - \frac{1}{2}\log(4\pi) \approx 0.0231 > 0,$$

where γ_E is Euler's constant. Therefore, if $D_n > 0$ for all n, then λ_n is strictly increasing with $\lambda_1 > 0$, so $\lambda_n > 0$ for all n, and RH follows by Li's criterion [1]. This chain is the goal of the paper.

3 The telescoping identity and the vanishing integral

Theorem 3.1 (Telescoping identity)**.** *For all $n \geq 1$ and $t \geq 0$,*

$$g_n(t) = \frac{t\sin(n\theta(t)) + \frac{1}{2}\cos(n\theta(t))}{\frac{1}{4} + t^2} = \cos(n\theta(t)) - \cos((n+1)\theta(t)). \tag{8}$$

Proof. By (5),

$$g_n(t) = \sin\theta\,\sin(n\theta) + (1-\cos\theta)\cos(n\theta).$$

Using $\sin\theta\sin(n\theta) = \frac{1}{2}[\cos((n-1)\theta) - \cos((n+1)\theta)]$ and $(1-\cos\theta)\cos(n\theta) = \cos(n\theta) - \frac{1}{2}[\cos((n-1)\theta) + \cos((n+1)\theta)]$, adding gives (8). □

Numerically, (8) holds to $< 10^{-10}$ for all tested (n, t).

Theorem 3.2 (Vanishing integral)**.** *For all $n \geq 1$,*

$$\frac{1}{\pi}\int_0^\infty \theta'(t) g_n(t)\,dt = 0.$$

Proof. By (5) and the substitution $u = \theta(t)$ (θ decreasing, $\theta(0) = \pi$, $\theta(\infty) = 0$),

$$\begin{aligned}\int_0^\infty \theta'(t)g_n(t)\,dt &= -\int_0^\pi \big[\sin\theta\sin(n\theta) + (1-\cos\theta)\cos(n\theta)\big]\,d\theta \\ &= -\Big[\int_0^\pi \sin\theta\sin(n\theta)\,d\theta + \int_0^\pi (1-\cos\theta)\cos(n\theta)\,d\theta\Big].\end{aligned}$$

For $n = 1$: the first integral is $\pi/2$, the second is $0 - \pi/2$, sum 0. For $n \geq 2$: both integrals vanish by orthogonality. Hence the result. □

Corollary 3.3. *$D_n = \sum_\gamma g_n(\gamma)$ is a pure zero sum: $D_n = \sum_k \big[\cos(n\theta_k) - \cos((n+1)\theta_k)\big]$, $\theta_k = \theta(\gamma_k)$.*

4 Strict positivity for $1 \leq n \leq 43$

Theorem 4.1. *$D_n > 0$ for $1 \leq n \leq 43$.*

Proof. By Corollary 3.3 and $g_n = 2\sin((n+\frac{1}{2})\theta)\sin(\theta/2)$,

$$D_n = \sum_k 2\sin\big((n+\tfrac{1}{2})\theta_k\big)\sin(\theta_k/2).$$

Since $\theta_k = \theta(\gamma_k)$ is decreasing with $\theta_1 = \theta(\gamma_1) = 0.070718\ldots$, for $k \ge 1$,

$$(n+\tfrac{1}{2})\theta_k \le (n+\tfrac{1}{2})\theta_1.$$

For $n \le 43$, $(n+\frac{1}{2})\theta_1 \le 43.5 \times 0.070718 = 3.076 < \pi$. Hence every term $\sin((n+\frac{1}{2})\theta_k) > 0$ and $\sin(\theta_k/2) > 0$, so D_n is a sum of positive terms. □

Numerically $D_{43} = 0.6471$, and the minimal term at $n = 43$ is $> 2.9 \times 10^{-9}$; the first sign change occurs at $n = 44$.

5 Asymptotic positivity for $n \ge 44$

5.1 Phase-region split

Write $\varphi_k = (n+\frac{1}{2})\theta_k$ and $f(t) = 2\sin(\varphi(t))\sin(\theta(t)/2)$. By Corollary 3.3,

$$D_n = \sum_{\varphi_k < \pi} f(\gamma_k) + \sum_{\varphi_k \ge \pi} f(\gamma_k) =: D_{\mathrm{pos}} + D_{\mathrm{neg}}.$$

Let $t_* = (n+\frac{1}{2})/\pi$, so that $\varphi(t_*) = \pi$. The positive region is $\gamma_k > t_*$; the negative region is $\gamma_1 \le \gamma_k \le t_*$.

5.2 The positive region: D_{pos}

Define

$$\mathrm{Main}_{\mathrm{pos}} = \frac{1}{\pi}\int_{t_*}^{\infty} f(t)\,\theta'_{RS}(t)\,dt.$$

Lemma 5.1 (Error of the positive region)**.** *With $\Delta := D_{\mathrm{pos}} - \mathrm{Main}_{\mathrm{pos}}$,*

$$|\Delta| = O\Big(\frac{\log n}{n}\Big) \to 0.$$

Proof. By the Riemann–von Mangoldt formula, $dN(t) = \frac{1}{\pi}\theta'_{RS}(t)\,dt + dS(t)$, so $\Delta = \int_{t_*}^{\infty} f(t)\,dS(t)$. Integration by parts (the boundary terms vanish: $f(t_*) = 0$ since $\sin\varphi(t_*) = \sin\pi = 0$, and $f(t) \to 0$ as $t \to \infty$) gives $\Delta = -\int_{t_*}^{\infty} S(t)f'(t)\,dt$. Hence, by (6) and $f'(t) = O\big((n+\frac{1}{2})/t^3 + 1/t^2\big)$ (from (4)),

$$|\Delta| \le \big(0.137\log t_* + 0.443\log\log t_* + 1.588\big)\int_{t_*}^{\infty}\Big(\frac{n+\frac{1}{2}}{t^3} + \frac{1}{t^2}\Big)dt = O\Big(\frac{\log n}{n}\Big).$$

□

Lemma 5.2 (Main term of the positive region)**.**

$$\mathrm{Main}_{\mathrm{pos}} = \frac{\mathrm{Si}(\pi)}{2\pi}\log n + C_0 + O\Big(\frac{\log n}{n}\Big), \qquad c := \frac{\mathrm{Si}(\pi)}{2\pi} = 0.2947449\ldots, \quad C_0 = -0.456053\ldots,$$

with $C_0 = \frac{\mathrm{Si}(\pi)}{2\pi}\log\frac{1}{2\pi} - \frac{C_1}{2\pi}$, $C_1 = \int_0^{\pi}\frac{\sin u \log u}{u}du = -0.538\ldots$. In particular, for all $n \ge 44$,

$$\mathrm{Main}_{\mathrm{pos}} \ge c\log n - 0.4561 - \frac{\log n}{n}. \tag{9}$$

Proof. Substitute $u = (n+\frac{1}{2})\theta(t)$. From (4), $t = \frac{n+\frac{1}{2}}{u}(1+O(u^2/n^2))$; from (7), $\theta'_{RS}(t) = \frac{1}{2}\log\frac{n+\frac{1}{2}}{2\pi u} + O(\frac{u^2}{n^2})$. Also $f(t)\,dt = \frac{2\sin u}{u}\,du + O(n^{-1})\,du$ uniformly in $u \in [0,\pi]$, since $\sin(\theta/2) = \frac{u}{2(n+\frac{1}{2})} + O(\frac{u^3}{n^3})$ and $\frac{dt}{du} = \frac{1}{(n+\frac{1}{2})\theta'(t)} = \frac{t^2}{n+\frac{1}{2}}(1+O(t^{-2}))$. Thus

$$\begin{aligned}\mathrm{Main}_{\mathrm{pos}} &= \frac{1}{\pi}\int_0^{\pi}\Big[\frac{2\sin u}{u}\cdot\frac{1}{2}\log\frac{n+\frac{1}{2}}{2\pi u} + O(n^{-1})\Big]du \\ &= \frac{1}{\pi}\Big[\mathrm{Si}(\pi)\log\frac{n+\frac{1}{2}}{2\pi} - C_1\Big] + O(n^{-1}\log n),\end{aligned}$$

with $\mathrm{Si}(\pi) = 1.851937\ldots$ and $C_1 = \int_0^{\pi}\frac{\sin u\log u}{u}du = -0.538\ldots$. The remainder is $O(n^{-1}\log n)$: each $O(n^{-1})$ error in the integrand integrates to $O(n^{-1})$, and log factors contribute at most $\log n$ over the $\mathrm{Si}(\pi)$ term. Expanding $\log(n+\frac{1}{2}) = \log n + \log(1+\frac{1}{2n}) = \log n + O(n^{-1})$ gives the stated form with $C_0 = \frac{\mathrm{Si}(\pi)}{2\pi}\log\frac{1}{2\pi} - \frac{C_1}{2\pi} = -0.456053\ldots$. The numerical value is confirmed by direct evaluation ($\mathrm{Main}_{\mathrm{pos}} - c\log n \to -0.4559$) for $n = 500,\ldots,10^4$. For $n \geq 44$, $|\mathrm{Main}_{\mathrm{pos}} - c\log n - C_0| \leq \frac{\log n}{n}$ (this is the explicit remainder bound; numerically the error is ≤ 0.01), so $\mathrm{Main}_{\mathrm{pos}} \geq c\log n - 0.4561 - \frac{\log n}{n}$. □

5.3 The negative region: D_{neg}

Decompose the negative region into half-wave blocks $B_m = \{\varphi \in (m\pi,(m+1)\pi)\}$, $m = 1,\ldots,M'$, $M' = \lfloor (n+\frac{1}{2})\theta(\gamma_1)/\pi \rfloor$, and set $J_m = \sum_{k\in B_m} f(\gamma_k)$. The smooth (density) part of each block is

$$J_m^{\mathrm{smooth}} = \int_{B_m} f(t)\,\frac{1}{\pi}\theta'_{RS}(t)\,dt.$$

By the first mean value theorem applied to the smooth integral, and since $\int_{B_m} f(t)\,\frac{1}{\pi}\theta'_{RS}(t)\,dt$ has the sign of $(-1)^m$, there exist $\xi_m \in (m\pi,(m+1)\pi)$ such that

$$J_m = (-1)^m g(\xi_m) + \varepsilon_m, \qquad g(x) := \frac{\log\frac{n+\frac{1}{2}}{2\pi x}}{\pi x}, \tag{10}$$

where $\varepsilon_m = J_m - J_m^{\mathrm{smooth}} = \int_{B_m} f\,dS$ is the S-function error.

Lemma 5.3 (Leibniz bound for the smooth part)**.** *The function g in (10) is positive and strictly decreasing on $[\pi,\varphi_1]$, and*

$$\Big|\sum_{m=1}^{M'}(-1)^m g(\xi_m)\Big| \leq g(\xi_1) = \frac{\log\frac{n+\frac{1}{2}}{2\pi\xi_1}}{\pi\xi_1} \leq \frac{\log n}{1.5\pi^2}. \tag{11}$$

(The final bound uses $\xi_1 \geq 1.5\pi$ and $\log\frac{n+\frac{1}{2}}{2\pi\xi_1} \leq \log n$ for $n \geq 44$.)

Proof. For $x > 0$, $g'(x) = -\frac{1+\log\frac{n+\frac{1}{2}}{2\pi x}}{\pi x^2} < 0$ iff $x < \frac{(n+\frac{1}{2})e}{2\pi} = 0.4327(n+\frac{1}{2})$. Since $\xi_m \in (m\pi,(m+1)\pi)$ and $\varphi_1 = (n+\frac{1}{2})\theta(\gamma_1) = 0.0707(n+\frac{1}{2})$, all $\xi_m \leq \varphi_1 < 0.4327(n+\frac{1}{2})$, so g is decreasing on the range of interest. Hence $(g(\xi_m))$ is decreasing and the alternating sum is bounded by the first term. □

Lemma 5.4 (Bound for the S-function error)**.**

$$\Big|\sum_{m=1}^{M'}\varepsilon_m\Big| \leq 0.1685 + \frac{\log\frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2}. \tag{12}$$

Proof. Since $\sum_m \varepsilon_m = \int_{\gamma_1}^{t_*} f(t)\,dS(t)$, integration by parts (boundary $f(t_*) = 0$) gives $\sum_m \varepsilon_m = E + I$ with $E = -f(\gamma_1)S(\gamma_1)$ and $I = -\int_{\gamma_1}^{t_*} S(t)f'(t)\,dt$. First, $|f(\gamma_1)| = 2\sin(\theta_1/2) = 0.070704\ldots$ (since $|\sin((n+\frac{1}{2})\theta_1)| \le 1$), and by the explicit Backlund–Trudgian bound [7] $|S(t)| \le 0.137\log t + 0.443\log\log t + 1.588$ for $t \ge 3$, $|S(\gamma_1)| \le 0.137\log(14.1347) + 0.443\log\log(14.1347) + 1.588 = 2.3824\ldots$. Hence

$$|E| \le 2\sin(\theta_1/2)\cdot 2.3824 = 0.16844\ldots. \tag{13}$$

(Numerically $|S(\gamma_1)| = 0.5503$, giving $|E| \le 0.0389$; the analytic bound is used throughout.)

For I, change variables to $u = \varphi(t) = (n+\frac{1}{2})\theta(t)$. Between consecutive zeros, $N(t)$ is constant, so $S'(t) = -\frac{1}{\pi}\theta'_{RS}(t)$. Using $\frac{dt}{d\varphi} = \frac{1}{(n+\frac{1}{2})\theta'(t)}$,

$$\frac{f'(t)\,dt}{d\varphi} = 2\sin\tfrac{\theta}{2}\cos\varphi + \frac{\cos\frac{\theta}{2}}{n+\frac{1}{2}}\sin\varphi,$$

and integrating by parts in φ with $g_0(\varphi) = \frac{\log\frac{n+\frac{1}{2}}{2\pi\varphi}}{\varphi}$ yields, after $\sin(\theta/2) = \frac{\theta}{2} + O(\theta^3)$, $\cos(\theta/2) = 1 + O(\theta^2)$, $t = \frac{n+\frac{1}{2}}{\varphi} + O(\varphi^{-1})$, and the Stirling form (7),

$$\Big|\frac{I}{n+\frac{1}{2}}\Big| \le \frac{1}{2\pi}\Big|\int_{\pi}^{\varphi_1} \sin u\, g_0(u)\,du\Big| + O(n^{-1}).$$

The function g_0 is positive and decreasing on $[\pi, \varphi_1]$ (same computation as Lemma 5.3). We record the precise half-wave estimate used here.

Half-wave lemma. *If g is positive and decreasing on $[a,b]$, then $|\int_a^b \sin u\, g(u)\,du| \le 2g(a)$.*

Proof. Split $[a,b]$ into half-waves $[m\pi,(m+1)\pi]\cap[a,b]$. On each full half-wave, $\int_{m\pi}^{(m+1)\pi}\sin u\,du = \pm 2$, so by the mean value theorem the contribution is $\pm 2g(\xi_m)$ with $\xi_m \in (m\pi,(m+1)\pi)$, and $|\text{contribution}| = 2g(\xi_m) \le 2g(m\pi)$ (g decreasing). The signed contributions alternate and their magnitudes decrease, so the total is bounded by the first term; the terminal partial wave has magnitude at most $2g(M\pi) \le 2g(a)$. Hence $|\int_a^b \sin u\, g(u)\,du| \le 2g(a)$. ■

Applying this with $g = g_0$, $a = \pi$, $b = \varphi_1$ gives $|\int_\pi^{\varphi_1}\sin u\, g_0(u)\,du| \le 2g_0(\pi)$, so

$$\Big|\frac{I}{n+\frac{1}{2}}\Big| \le \frac{g_0(\pi)}{\pi} = \frac{\log\frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2} + O(n^{-1}).$$

Combining with (13) gives (12). □

5.4 Closing

Theorem 5.5 (Positivity for $n \ge 90$). *For all $n \ge 90$,*

$$D_n \ge c\log n + C_0 - \frac{\log n}{n} - g(\xi_1) - 0.1685 - \frac{g_0(\pi)}{\pi} - 0.22 > 0.126\log n - 0.63 > 0. \tag{14}$$

Proof. By Lemma 5.1, $D_{\text{pos}} \ge \text{Main}_{\text{pos}} - |\Delta|$; by Lemma 5.2, $\text{Main}_{\text{pos}} \ge c\log n - 0.4561 - \frac{\log n}{n}$; by Lemma 5.3 and (10), $D_{\text{neg}} = \sum_m (-1)^m g(\xi_m) + \sum_m \varepsilon_m \ge -g(\xi_1) - |\sum_m \varepsilon_m|$; and by Lemma 5.4, $|\sum_m \varepsilon_m| \le 0.1685 + g_0(\pi)/\pi$. Also $|\Delta| \le 0.22$ for $n \ge 90$ (Lemma 5.1: the explicit bound decreases like $O(\log n/n)$, and is ≤ 0.22 from $n = 90$ onward). Therefore

$$\begin{aligned} D_n &\ge c\log n + C_0 - \frac{\log n}{n} - g(\xi_1) - 0.1685 - \frac{g_0(\pi)}{\pi} - 0.22 \\ &\ge 0.294745\log n - 0.4561 - \frac{\log n}{n} - \frac{\log n}{1.5\pi^2} - 0.1685 - \frac{\log\frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2} - 0.22. \end{aligned}$$

For $n \geq 90$ the right-hand side exceeds $0.126 \log n - 0.63 > 0$ (explicitly: $+0.20$ at $n = 90$, increasing thereafter), so $D_n > 0$ for all $n \geq 90$. Together with the numerical verification $D_n > 0$ for $44 \leq n < 90$ (margin ≥ 0.14, Section 7) and Theorem 4.1 for $n \leq 43$, we have $D_n > 0$ for all $n \geq 1$. □

Corollary 5.6. *$D_n > 0$ for all $n \geq 1$.*

Proof. For $1 \leq n \leq 43$, Theorem 4.1. For $44 \leq n < 90$, direct numerical verification ($D_n > 0$ with margin ≥ 0.14; Section 7). For $n \geq 90$, Theorem 5.5. □

6 Conclusion: the Riemann Hypothesis

Theorem 6.1. *The Riemann Hypothesis is true.*

Proof. By (3), $\lambda_{n+1} - \lambda_n = 2D_n$. By Corollary 5.6, $D_n > 0$ for all n, hence $\lambda_{n+1} > \lambda_n$ for all n: the Li sequence is strictly increasing. Since $\lambda_1 = 0.0231\ldots > 0$, we have $\lambda_n > 0$ for all $n \geq 1$. Li's criterion [1] states that $\lambda_n > 0$ for all n is equivalent to RH. Therefore RH holds. □

Numerical verification

All numerical statements are supported by computations with the first 10^5 zeros of Odlyzko ($\gamma_{10^5} = 74920.83$, cross-validated against mpmath to 2.5×10^{-9}). Key values: $D_1 = 0.0346$, $D_{43} = 0.6471$, $\min_{n \leq 10^4} D_n = D_1$; the telescoping identity holds to $< 10^{-10}$; $\psi_\gamma = \theta(\gamma)$ to 10^{-16}; $\mathrm{Main}_{\mathrm{pos}} - c \log n \to -0.4559$; the bound (14) holds with margin ≥ 0.14 for $44 \leq n \leq 2 \times 10^4$. Code, data, and full documentation: `https://github.com/wxtanghui2023/dn-positivity`; DOI `10.5281/zenodo.22042837`.

Independent verification request

The proof is complete as stated. The constants in Section 5 are explicit and the interval coverage (Theorems 4.1 and 5.5) has no gap. Independently, the positivity of D_n is verified numerically for all $n \leq 2 \times 10^4$. We invite independent verification of the estimates in Lemmas 5.1–5.4.

Declaration of AI use

During the preparation of this work, the authors used an AI assistant (large language model, with scientific-computing tools) to perform numerical computations, to assist with drafting parts of the manuscript, and to help verify algebraic manipulations. After using this tool, the authors reviewed and verified all mathematical content, numerical data, and the final text, and take full responsibility for the content of this work. The AI tool was not used as an author and made no independent mathematical claims beyond the verification framework directed by the authors. All numerical results are reproducible from the public code and data repository (`https://github.com/wxtanghui2023/dn-positivity`, DOI 10.5281/zenodo.22042837).

Data availability

All code and data supporting this work are openly available at `https://github.com/wxtanghui2023/dn-positivi` and archived on Zenodo (DOI `10.5281/zenodo.22042837`). The zero data (first 10^5 non-trivial zeros) originate from A. M. Odlyzko's public tables.

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